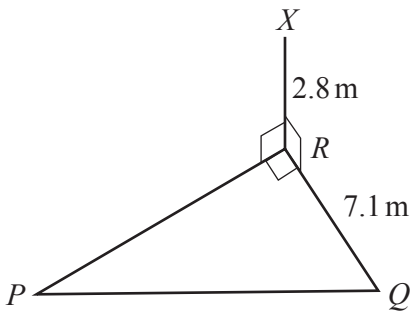
0580_w23_qp_42_Q7

7 **(a)**



NOT TO SCALE

The diagram shows a right-angled triangle *PQR* on horizontal ground.
X is vertically above *R* and the angle of elevation of *X* from *P* is 21°.
XR = 2.8 m and *RQ* = 7.1 m.

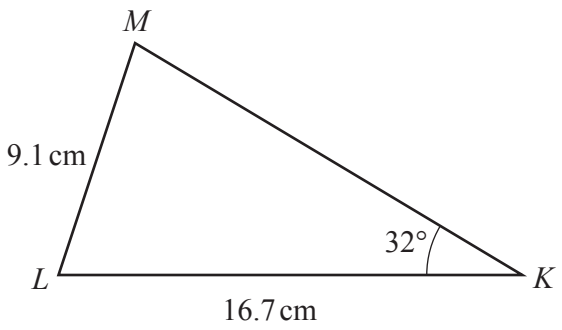
(i) Calculate the angle of elevation of *X* from *Q*.

..... [2]

(ii) Calculate *PQ*.

..... m [3]

(b)



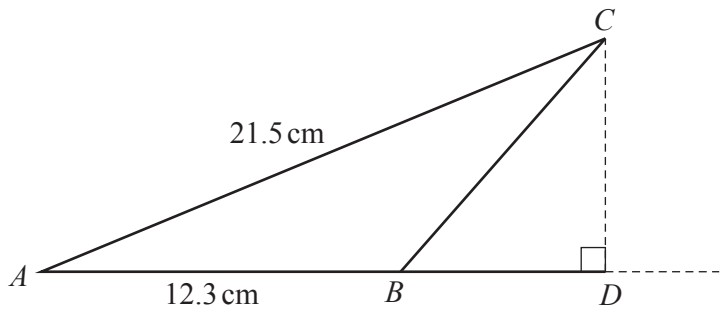
NOT TO SCALE

Calculate the acute angle *KML*.

Angle *KML* = ... [3]

0580_w23_qp_42_Q7

(c)



NOT TO SCALE

The area of triangle ABC is $62.89\,\text{cm}^2$.

(i) Show that angle $BAC = 28.4°$, correct to 1 decimal place.

[2]

(ii) Calculate BC.

..... cm [3]

(iii) AB is extended to a point D such that angle $BDC = 90°$.

Calculate BD.

..... cm [3]